

ContainIQ

AI-Powered Incident Response Platform

Customer Onboarding Lab Guide

Hands-on exercises for SOC analysts and security teams

Version 1.0

3 Hands-on Labs

~30 Min Total

containiq.io

Open-source | AI-native | Audit-ready | Free vs \$50k+/yr enterprise tools

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INTRODUCTION

Getting Started with ContainIQ

What is ContainIQ?

ContainIQ is an open-source, AI-native Security Orchestration, Automation, and Response (SOAR) platform built for SOC teams and MSSPs who need enterprise-grade incident response without the \$50,000+/year price tag of tools like CrowdStrike Falcon Fusion or Cortex XSOAR.

At its core, ContainIQ combines intelligent alert ingestion, AI-powered case analysis (via Claude), structured playbook execution with approval gates, NIST CSF 2.0 compliance mapping, IOC enrichment, and a full immutable audit trail — all in one unified platform.

Who This Guide Is For

- SOC Analysts who investigate and respond to security incidents
- MSSP engineers onboarding new clients onto ContainIQ
- Security engineers evaluating ContainIQ for their organization
- Anyone learning modern AI-assisted incident response workflows

Prerequisites

Requirement	Details
Python	3.11 or higher
ContainIQ	Running locally (python run.py) on port 5001, or hosted
Browser	Any modern browser (Chrome, Firefox, Edge)
curl	For Lab 1 API exercise (or use Postman / Python requests)
API Keys (optional)	VIRUSTOTAL_API_KEY, ABUSEIPDB_API_KEY for IOC enrichment
ANTHROPIC_API_KEY	Required for AI analysis features (Labs 2 & 3)

Lab Environment Overview

Each lab builds on the last. You will ingest an alert, investigate it with AI, contain it with a playbook, and document it for compliance — completing the full SOC workflow from detection to closure in about 30 minutes.

Lab	Scenario	Skills Practiced	Time
Lab 1	Phishing / Inbox Forwarding	Alert ingestion, IOC enrichment	~8 min
Lab 2	Ransomware Containment	AI analysis, playbook execution	~12 min
Lab 3	Incident Closure	Reporting, NIST CSF, MTTR metrics	~10 min

LAB 1

Alert Ingestion & Case Creation

SCENARIO: Your SIEM fires on a suspicious inbox forwarding rule created on the CEO's account. The login originated from a known TOR exit node and the rule forwards all email to an external domain. You need to get this into ContainIQ fast and start the investigation.

Step 1 — Start ContainIQ and Log In

Open your terminal and run: **python run.py** from the ContainIQ directory. Then navigate to **http://localhost:5001** and log in with: **admin@containiq.local / changeme**

Step 2 — Ingest the Simulated Alert via API

Open a new terminal and run the following curl command to simulate a Microsoft Sentinel alert firing on the inbox forwarding rule:

```
curl -X POST http://localhost:5001/api/v1/alerts \
-H "X-API-Key: your-ingest-key" \
-H "Content-Type: application/json" \
-d '{
  "title": "Suspicious Inbox Forwarding Rule Detected",
  "severity": "high",
  "source": "microsoft_sentinel",
  "user": "ceo@company.com",
  "src_ip": "185.220.101.45",
  "domain": "evil-exfil.ru",
  "description": "Inbox forwarding rule created to external address "
  "after suspicious login from TOR exit node"
}'
```

You should receive a JSON response with **status: created** and a **case_id**.

Step 3 — Navigate to the Created Case

In ContainIQ, click **Cases** in the nav bar. The new case will appear at the top with severity **HIGH**. Click **Investigate** to open it.

Step 4 — Review Auto-Extracted Entities

In the Entities panel on the right, you will see ContainIQ automatically extracted three IOCs from the alert payload:

- user: ceo@company.com
- ip: 185.220.101.45 (TOR exit node)
- domain: evil-exfil.ru (exfiltration domain)

Step 5 — Observe IOC Enrichment (if API keys configured)

If you have VIRUSTOTAL_API_KEY or ABUSEIPDB_API_KEY set in your .env file, enrichment runs automatically in the background. Within a few seconds, the entities will show color-coded threat badges:

Badge	Meaning
High Risk (red)	5+ VT detections OR AbuseIPDB score $\geq$ 75%
Suspicious (yellow)	1+ VT detections OR AbuseIPDB score $\geq$ 25%
Clean (green)	No detections found
VT X/N	VirusTotal: X malicious engines out of N total
Abuse XX%	AbuseIPDB confidence score
TOR	IP is a known TOR exit node

Expected Outcome: Case created automatically with severity HIGH. Three entities extracted (user, IP, domain). IOC enrichment badges visible if API keys are configured. Timeline shows 'alert_fired' event.

What Just Happened?

ContainIQ's ingestion API auto-detected the alert as a generic JSON payload, extracted IOC entities from known field names (src_ip, user, domain), created a case record, logged an immutable audit entry, and fired notifications for high-severity cases — all in under a second. The IOC enrichment runs in a background thread so it never slows down ingestion. Every action is recorded in the audit trail for compliance.

LAB 2

AI Investigation & Playbook Execution

SCENARIO: Ransomware execution is detected on workstation DESKTOP-4A2X. File encryption has started. You have minutes to contain it before it spreads to network shares. ContainIQ's AI analyst and automated playbooks will help you move at machine speed.

Step 1 — Create the Ransomware Case

Click **New Case** from the dashboard. Fill in the following details:

```
Title: Ransomware Execution — DESKTOP-4A2X
Severity: Critical
Source: crowdstrike
Description: Ransomware detected. File encryption in progress on DESKTOP-4A2X.
Entities (one per line):
device:DESKTOP-4A2X
hash:4d1740485713a2ab3a4f5822a01f645ff13dca5
ip:10.0.1.44
```

Step 2 — Run AI Analysis

Click the **AI Analyze** button in the case detail header. Claude will analyze the case and return:

- Attack summary and stage classification
- MITRE ATT&CK; technique mapping (e.g. T1486 Data Encrypted for Impact)
- Severity rationale
- Immediate recommended actions
- Investigation gaps to fill

Step 3 — Generate Threat Hunt Queries

In the **AI Threat Hunting** panel, type a question and select your SIEM:

```
Question: Hunt for lateral movement from 10.0.1.44 in the last 48 hours
SIEM: Splunk SPL
```

Claude will generate a ready-to-run SPL query you can paste directly into Splunk.

Step 4 — Run the Ransomware Playbook (Dry Run)

In the **Run a Playbook** panel, select **Ransomware Execution Response**. The playbook runner will open with **Dry Run** mode enabled by default.

Step 5 — Review Blast Radius and Approve Execution

Review the dry run output — it shows every action the playbook will take, which hosts it will touch, and the blast radius limit. When satisfied, click **Run Live** and approve each step that requires an approval gate. Steps include:

→ Isolate endpoint DESKTOP-4A2X from network

- Disable user account associated with the case
- Snapshot disk for forensic preservation
- Block the malicious hash across all endpoints

Step 6 — Verify Rollback Capability

In the playbook run detail, click **Rollback** to see the rollback plan. ContainIQ maintains a full rollback manifest so every automated action can be reversed if needed — critical for audit compliance.

Expected Outcome: Case analyzed with MITRE mapping. Splunk hunt query generated. Ransomware playbook executed with approval gates passed. Rollback manifest available. All steps logged in audit trail.

What Just Happened?

The AI analyst sent the full case context to Claude with prompt caching for efficiency. The playbook engine validated each step against the blast radius limit before executing. Approval gates ensure no automated action runs without human sign-off — a core requirement for SOC compliance frameworks. The rollback manifest is generated at run time so reversibility is guaranteed before the first action fires.

LAB 3

Reporting & Compliance

SCENARIO: The ransomware incident is contained. Your CISO needs a full incident report by end of day, the compliance team needs a NIST CSF 2.0 mapping, and your manager wants to know the MTTR for this quarter. ContainIQ generates all of this in minutes.

Step 1 — Close the Case

Open the ransomware case from Lab 2. Click **Set Status** and move it through: **Investigating** → **Contained** → **Closed**. ContainIQ automatically records the closed_at timestamp for MTTR calculation.

Step 2 — Generate an Incident Report

In the **Generate Report** panel on the case detail page, click **Incident Report**. The AI analyst produces a structured report including: executive summary, timeline of events, affected assets, containment actions taken, and recommendations.

Step 3 — Generate NIST CSF 2.0 Mapping

Click **NIST CSF Mapping**. ContainIQ maps every response action taken during the case to NIST CSF 2.0 functions: Govern (GV), Identify (ID), Protect (PR), Detect (DE), Respond (RS), Recover (RC). This is what compliance auditors want to see.

Step 4 — Export as PDF

Click **Export PDF**. ContainIQ generates a formatted PDF combining the incident report and NIST mapping — ready to send to your CISO or attach to a compliance submission.

Step 5 — Review the Immutable Audit Trail

Navigate to **Reports** → **Audit Packet**. Every action taken during the case is recorded with: timestamp, user, action type, outcome, and IP address. This log is append-only and cannot be modified — meeting SOC 2, ISO 27001, and NIST requirements.

Step 6 — Check MTTR on the Metrics Dashboard

Click **Metrics** in the nav bar. The MTTR dashboard shows:

- Overall average MTTR (hours) across all closed cases
- p50 and p95 response times
- Breakdown by severity (critical, high, medium, low)
- 8-week trend chart
- SLA benchmark comparison (e.g. critical cases: target 1h)
- Stale open cases breaching the 24h SLA threshold

Expected Outcome: Case closed with full timestamp. Incident report generated. NIST CSF 2.0 mapping produced. PDF exported. Immutable audit packet ready. MTTR recorded and visible in

metrics dashboard.

What Just Happened?

ContainIQ's NIST mapper analyzed every action in the case timeline and audit log, then classified each one against the six NIST CSF 2.0 functions automatically — no manual tagging required. The MTTR engine calculated response time from the moment the case was created to the moment it was closed, giving you real data for SLA reporting, team performance reviews, and compliance audits. The immutable audit log ensures you can prove — not just claim — what happened and when.

APPENDICES

Reference Material

Appendix A — API Quick Reference

Method	Endpoint	Description
POST	/api/v1/alerts	Ingest an alert (auto-detect format)
POST	/api/v1/alerts?format=sentinel	Ingest with explicit format override
POST	/api/v1/alerts/test	Validate payload without creating a case
GET	/api/v1/health	Health check — returns service status

Authentication Header:

```
X-API-Key: <your-INGEST_API_KEY>
```

Appendix B — Supported Alert Formats

Format	Auto-Detection Signal	format= value
Microsoft Sentinel	alertDisplayName in properties	sentinel
Splunk	search_name or result field present	splunk
CrowdStrike Falcon	behaviors or detection_id present	crowdstrike
Elastic SIEM	signal.rule nested object present	elastic
Generic JSON	All other payloads (always works)	generic

Appendix C — Environment Variables

Variable	Required?	Purpose
ANTHROPIC_API_KEY	Yes (AI features)	Powers AI analysis, hunt queries, threat intel synthesis
INGEST_API_KEY	Yes (API ingest)	Authenticates external alert sources
VIRUSTOTAL_API_KEY	Optional	IOC enrichment for IPs, domains, file hashes
ABUSEIPDB_API_KEY	Optional	IP reputation and abuse confidence scoring
SHODAN_API_KEY	Optional	Internet-exposed asset intelligence
OTX_API_KEY	Optional	AlienVault OTX threat intelligence feed

DATABASE_URL	Optional	PostgreSQL URI (defaults to SQLite for dev)
SECRET_KEY	Production	Flask session signing key — change in production

For the latest documentation, changelog, and community support: github.com/containiq | containiq.io/docs

